

SIBISANJAY B

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PROFESSIONAL SUMMARY

Aspiring ECE undergraduate seeking an opportunity in semiconductor industry to apply my knowledge in Digital Electronics, Verilog HDL, System Verilog, RTL Design, UVM and Embedded C. Hands-on experience in VLSI Design & Verification and passionate about semiconductor technology and eager to contribute to ASIC/FPGA design, functional verification and embedded system development while continuously enhancing my technical expertise.

ACADEMIC EDUCATION

B.E - ECE 2023-2027
KSR College of Engineering | Namakkal
Present CGPA: 7.76

HSC | Kongu Higher Secondary School | Karur 2022-2023
Percentage: 82%

SSLC | Kongu Higher Secondary School | Karur 2020- 2021
Percentage: 86.8%

PROJECTS

- **IoT Based Alcohol Detection**

Tech Stack: ESP32, Embedded C, MQ-3, LCD Display, Buzzer, GSM/Wi-Fi.
Problem Statement: Developed an IoT-based vehicle safety system to prevent accidents caused by drunk driving through real-time alcohol detection and instant alert generation.
Innovation: Integrated alcohol sensing with GSM/Wi-Fi notifications, buzzer alarms and ignition control logic for automated accident prevention.
Future Scope: Can be enhanced with facial recognition, cloud monitoring, GPS tracking and AI-based intoxication analysis for smart transportation systems.

- **Landslide Detection**

Tech Stack: Soil Moisture, Rain, Vibration Sensors, Embedded C, Arduino IDE, GSM Module, ESP32.
Problem Statement: Designed a real-time landslide early warning system to reduce loss of life and property by continuously monitoring environmental parameters.
Innovation: Combined multiple sensors with ESP32 and GSM technology to provide accurate multi-parameter analysis and instant warning alerts before potential landslides.
Future Scope: Can be integrated with IoT cloud platforms, machine learning prediction models, GPS mapping and government disaster management networks.

MINI PROJECTS

- **Time Divider using Verilog HDL**
  Designed and verified a clock frequency divider by 2 using Verilog HDL for digital system applications.
- **Traffic Light Controller System using Verilog HDL**
  Developed and simulated an FSM-based traffic signal controller using Verilog HDL for automated traffic management.

INTERNSHIPS

- Technical Trainee at BSNL | Salem — Aug 2024
- Embedded Systems Embedded C – AVR GPIO at Embuzz Technologies — Jun 2025
- Intern at Taras Systems and Solution | VLSI Design & Verification (6mon) — May 2026

CERTIFICATIONS

- NPTEL Certification | Cloud Computing - Elite 73% — Oct 2024
- MathWorks - Data Analysis Techniques| Regression with Deep Learning — Aug 2025
- NPTEL Certification | Industrial 4.0 And IoT - Elite Silver 75% — Apr 2026

PRESENTATION AND CONFERENCES

- International Conference on IOT based Alcohol Detection | KSRCE — Mar 2025
- Technova Hackathon | Landslide Detection Monitoring System — Mar 2025
- Chakravyuha 2025 National Level Tech Fest | Health Monitoring System — Oct 2025
- Smart Attendance using Face Recognition at CYBERZEC Karpagam Academy — Feb 2026

WORKSHOPS & VALUE-ADDED COURSES

- Introduction to Quantum Computing at K.S.Rangasamy College of Technology — Sep2025
- Hands-On Project-Based Workshop on PALS Thinkathon — Aug2024
- VAC on RTL Design using Verilog HDL | Functional Verification and UVM — Jan 2026
- VAC on PCB Designing | KSRCE — Dec 2025

SKILLS

Programming languages: Verilog HDL | System Verilog | Embedded C | UVM Verification | RTL Design | FFPGA Basics
Soft Skills: Strong Communication | Conflict Resolution | Problem Solving | Critical Thinking
Developer Tool: VS Code | Multisim Software | EDA Playground | Xilinx Vivado | VS Code